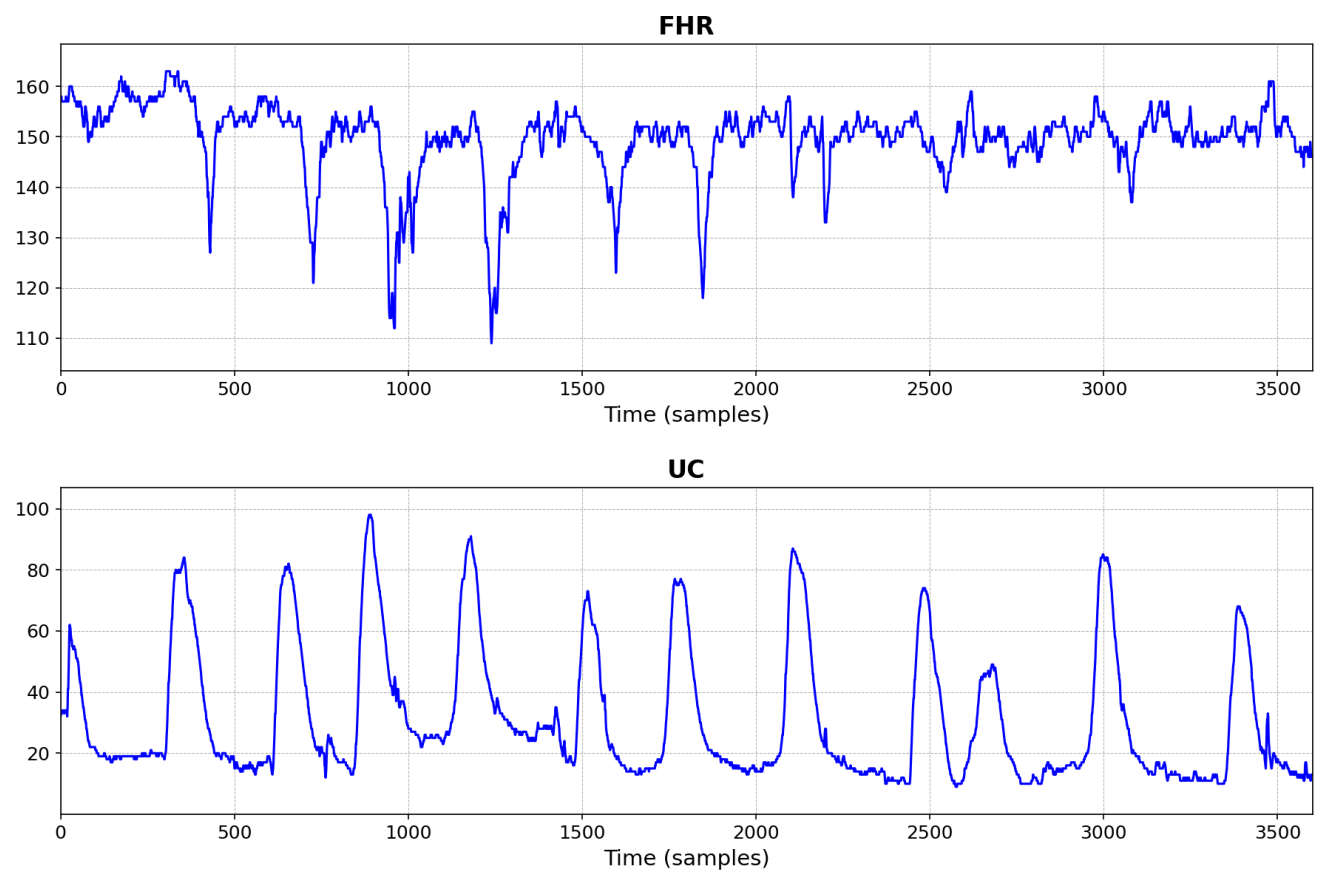
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# CTGDiff: A Conditional Diffusion Model for Cardiotocography Signal Synthesis

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Reporter: Pufan Cai

# 1 Background and Motivation




## Reflect Fetal Health

Cardiotocography(CTG) is an important indicator to evaluate the health status of the fetus.

## The composition of CTG

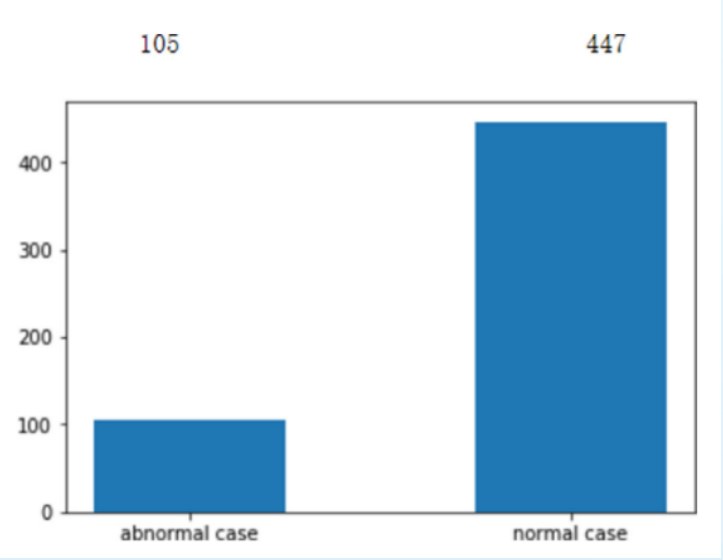
Each CTG data record contains two signals:the fetal heart rate and the uterine contraction signal.

# CTU-CHB Intrapartum Cardiotocography Database

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**CTU-UHB Intrapartum Cardiotocography Database** *(Feb. 18, 2014, 11:30 a.m.)*

This collection of 552 CTGs from the Czech Technical University (CTU) in Prague and the University Hospital in Brno (UHB) was carefully selected from 9164 recordings recently collected at UHB. Each recording is up to 90 minutes long, and includes a fetal heart rate time series, a uterine contraction signal, and maternal, delivery, and fetal clinical details.




At present, the most commonly used publicly available dataset is CTU-CHB, however, this dataset has only 552 data sets, and the data categories are extremely unbalanced.

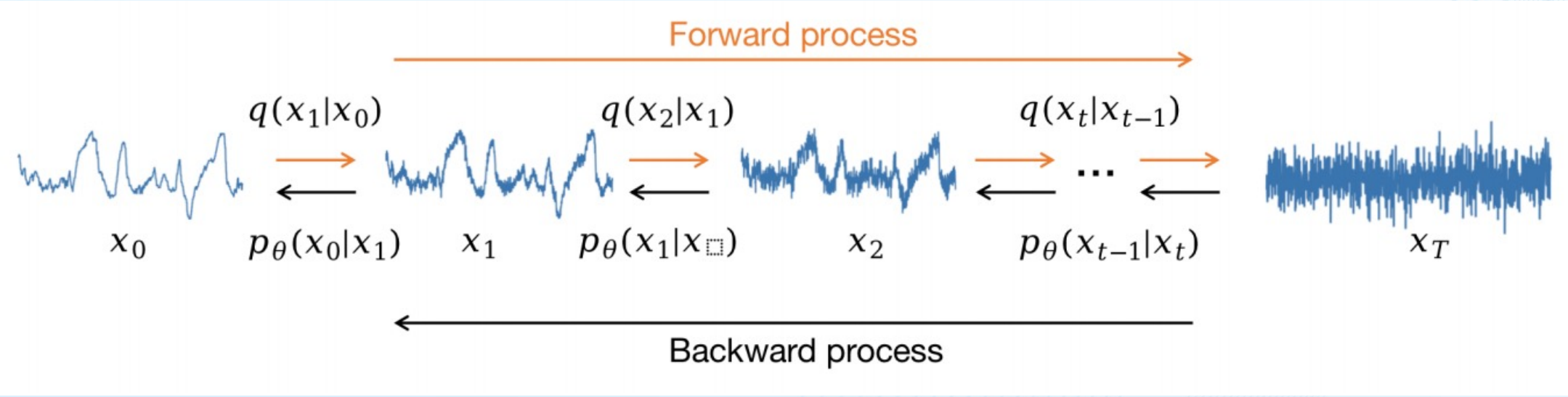
As shown in the figure below on the left, among the 552 ctg data, 447 were normal data and only 105 were pathological data

# 2 Method

## 2.1 Diffusion Probabilistic Models

Diffusion models, also known as diffusion probabilistic models, are a class of latent variable generative models.

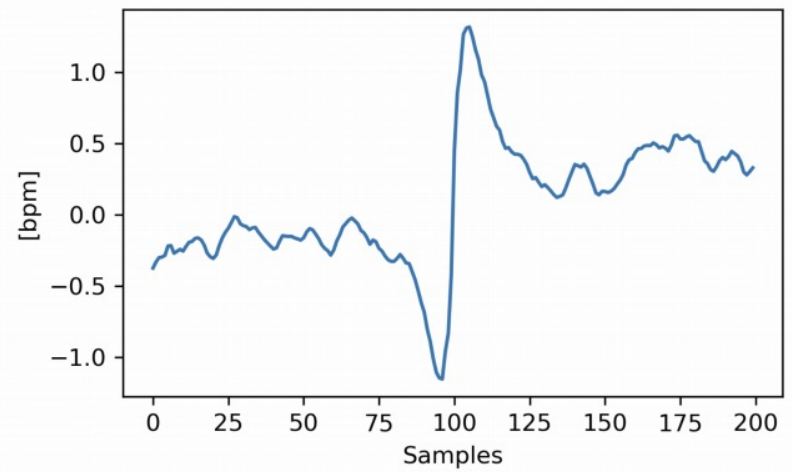
Due to its success in the field of image generation, some work began to apply the Diffusion model to the field of signal generation.Additionally, there has been no prior work on applying Diffusion model to CTG signal generation.



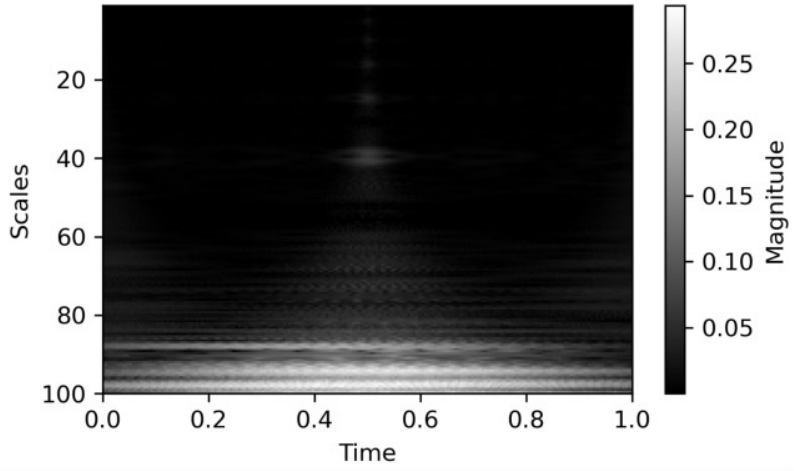
## 2.2 PRSA Curve and Spectrogram

We use the Phase-Rectified Signal Averaging (PRSA) spectrogram as the label for each signal.

The spectral method derived from PRSA better differentiates high-risk pregnancies compared to other traditional spectrum estimators. We compute the spectrogram from the wavelet coefficients. The PRSA curve and its spectrogram are shown below.

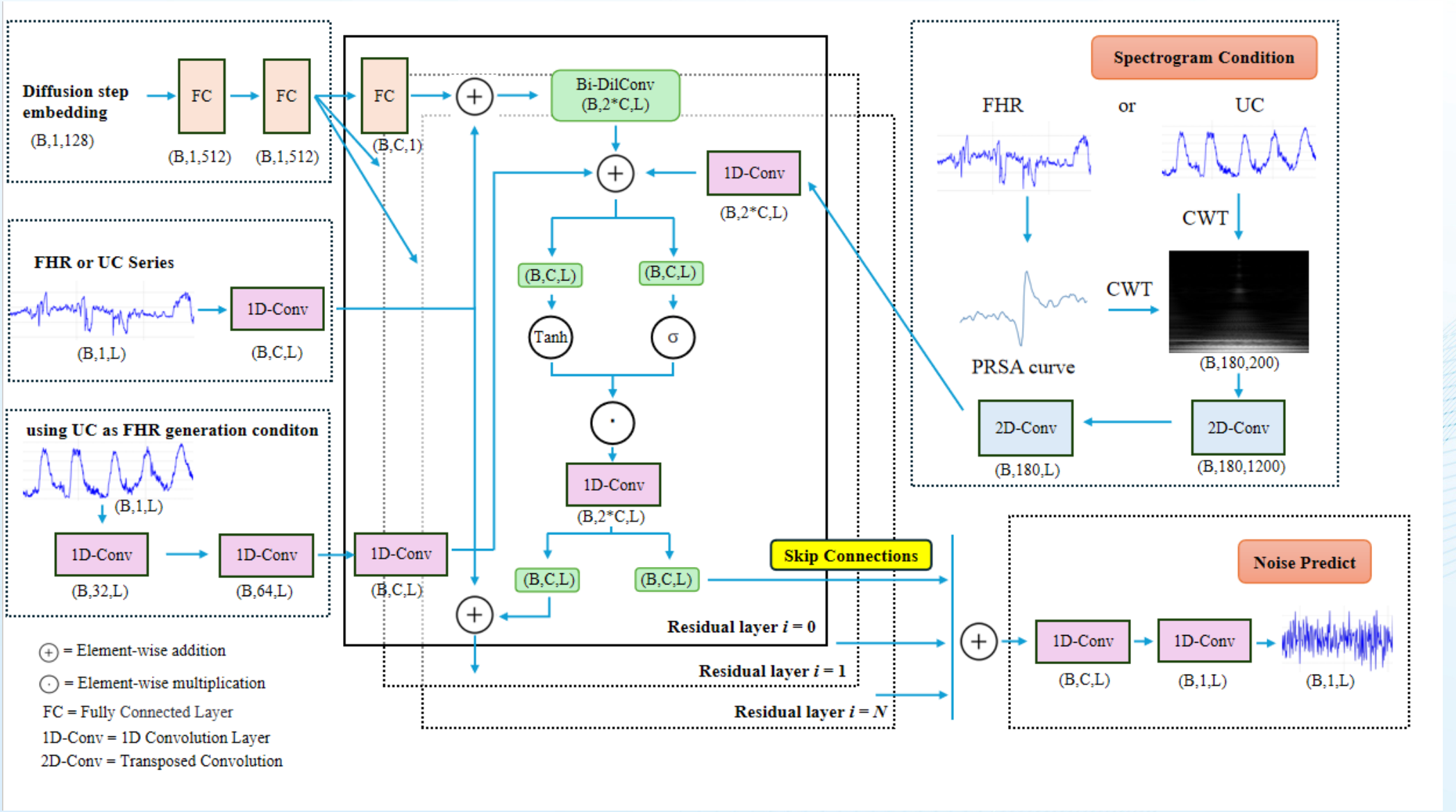



(a) PRSA curve of FHR series.




(b) CWT of PRSA data.

## 2.3 Model Architecture

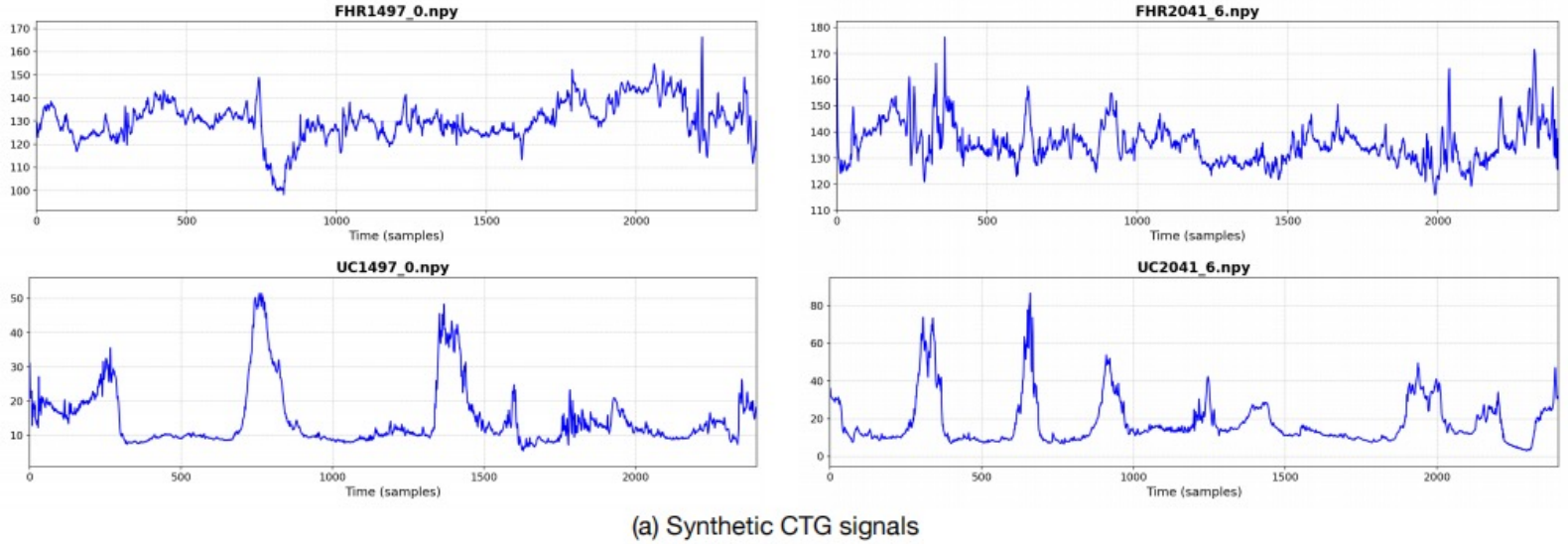


# 3 Result Analysis

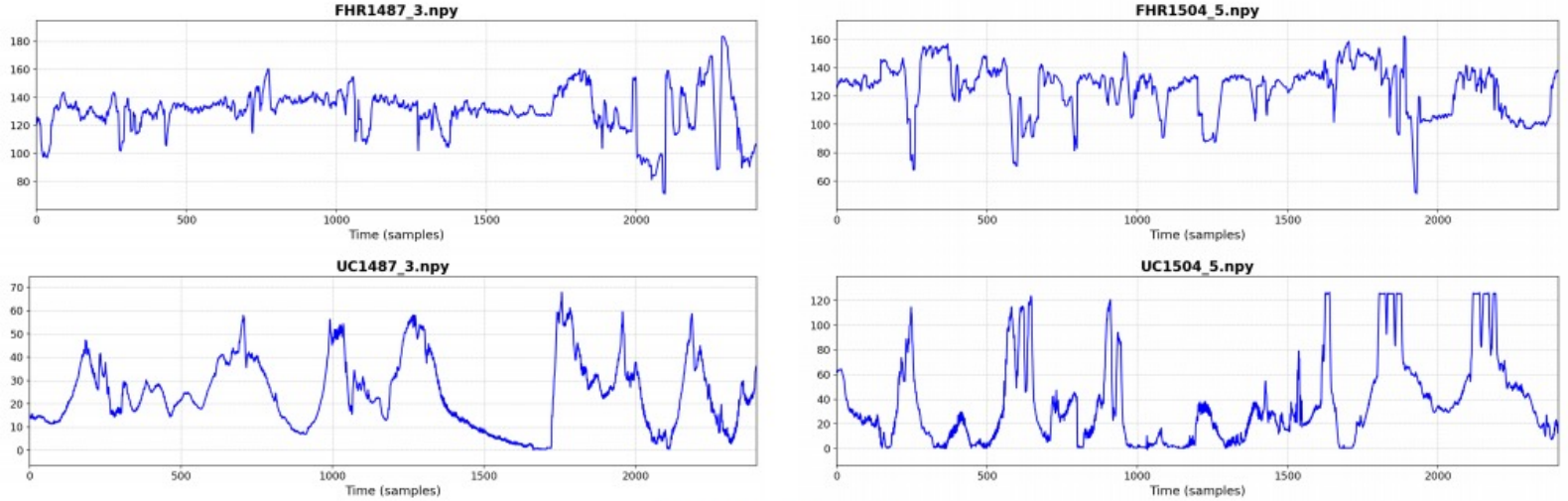
**Comparison of Different Models for Synthesis**

| Model | Evaluation Metrics | | | | |
|---|---|---|---|---|---|
| | RE(↓) | FD(↓) | MMD(↓) | SWD(↓) | PRD(↓) |
| SeqGAN | 10.324 | 0.974 | 6.527 | 0.541 | 10.635 |
| DCGAN | 9.532 | 0.795 | 5.267 | 0.254 | 10.431 |
| CGAN | 9.012 | 0.763 | 5.022 | 0.254 | 9.431 |
| CWGAN | 8.523 | 0.715 | 4.568 | 0.173 | 0.702 |
| FHRGAN | 8.059 | 0.682 | 1.975 | 0.086 | 5.437 |
| CTGGAN | 5.621 | 0.614 | 0.215 | **0.012** | 4.821 |
| DDPM | 7.059 | 0.682 | 6.323 | 0.521 | 9.786 |
| **CTGDiff** | **3.951** | **0.291** | **0.004** | 0.245 | **0.646** |

# Comparison of Synthetic Signal and Real CTG Signal



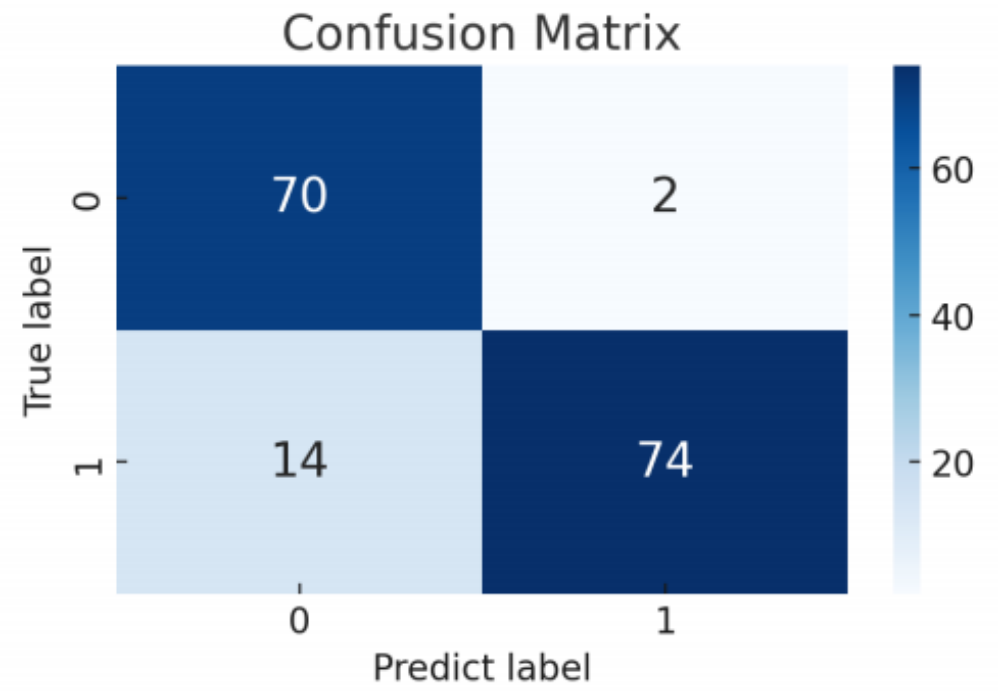

(a) Synthetic CTG signals




(b) Real CTG signals

As the figure shows, CTGDiff can generate high-fidelity synthetic CTG signals that closely align with real world data.

**obstetrician assessment**




we included evaluations of 160 synthetic samples by three obstetricians in our study.We extracted PRSA curve spectrograms from the test set, which served as conditions for our trained model to generate diverse data.

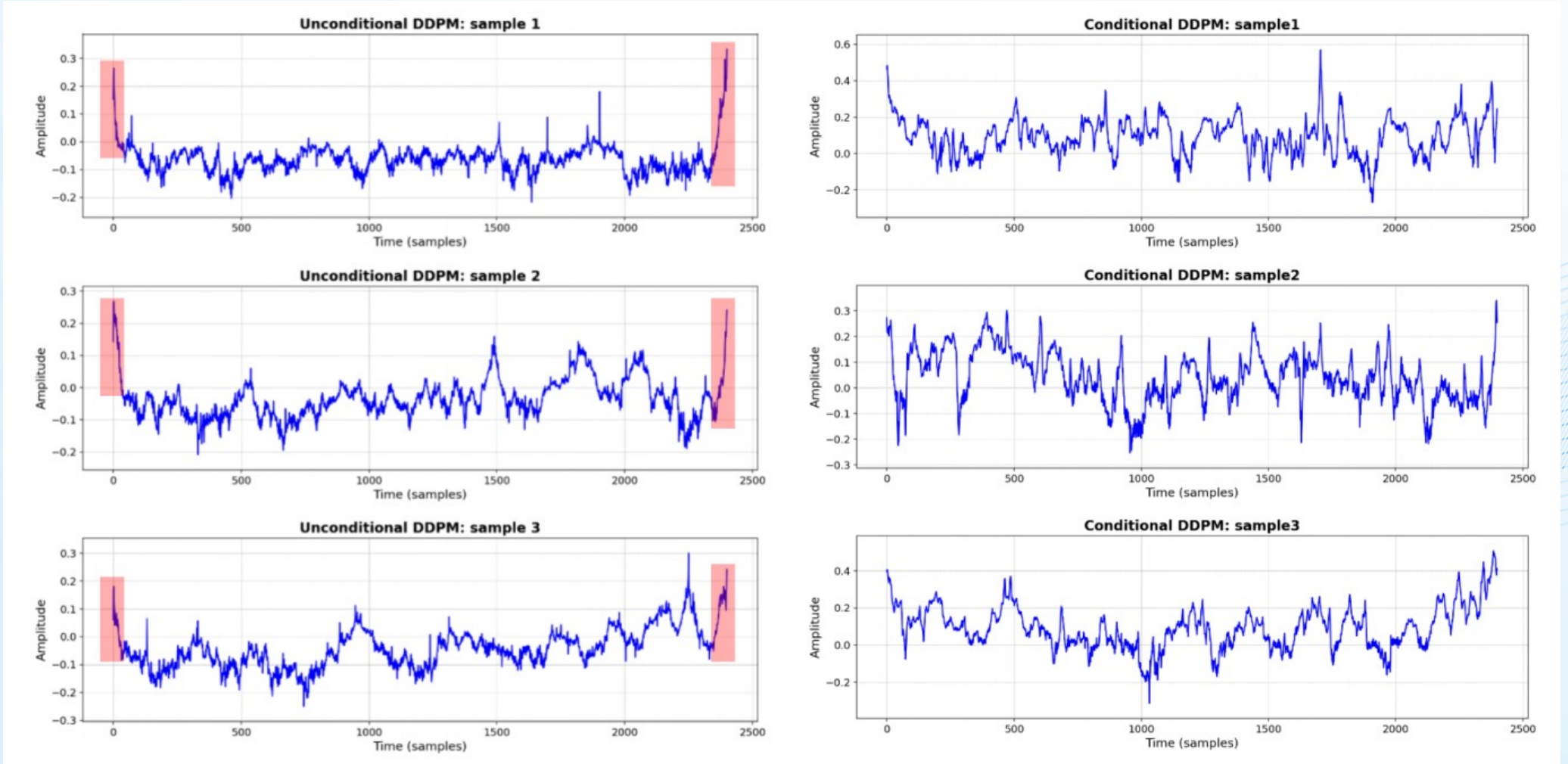
## Classification Accuracy

To verify the quality of the generated signals, we trained a classification model designed to distinguish between normal and abnormal FHR signals. We mixed both real and generated signals for training the classification model, and then evaluated the classification accuracy on real signals.

| No. | Real Signal | Synthetic Signal | Accuracy(↑) |
|---|---|---|---|
| 1 | 100% | 0% | 75.0% |
| 2 | 90% | 10% | 73.0% |
| 3 | 80% | 20% | 73.0% |
| 4 | 70% | 30% | 69.0% |
| 5 | 60% | 40% | 67.1% |
| 6 | 50% | 50% | 68.5% |

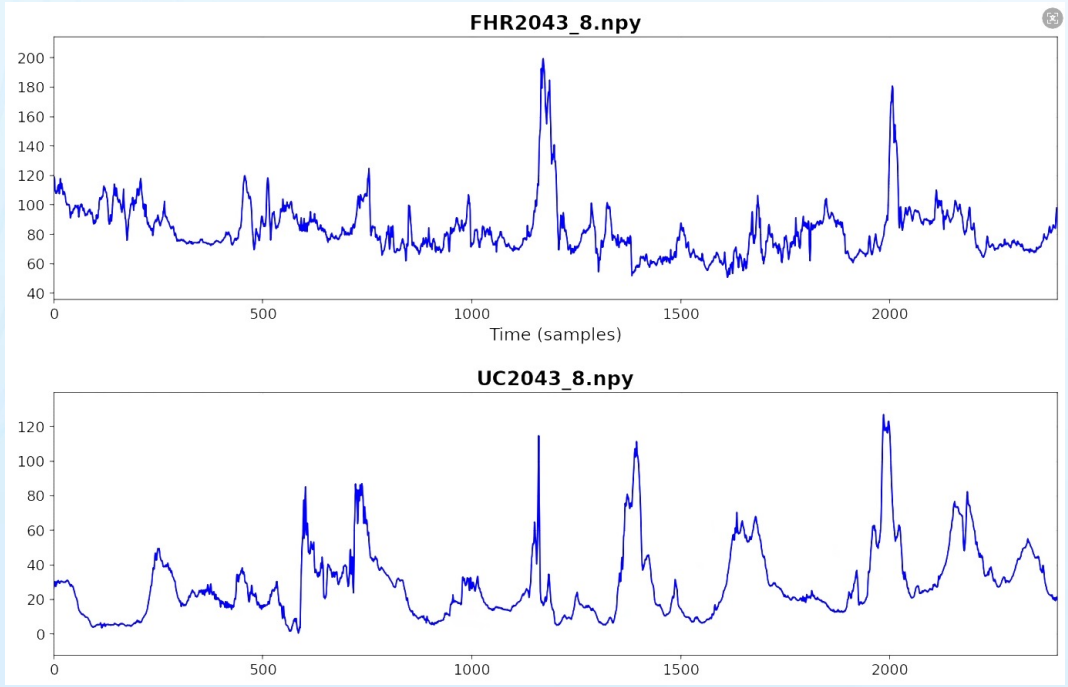
# Evaluation of Conditional and Unconditional Models

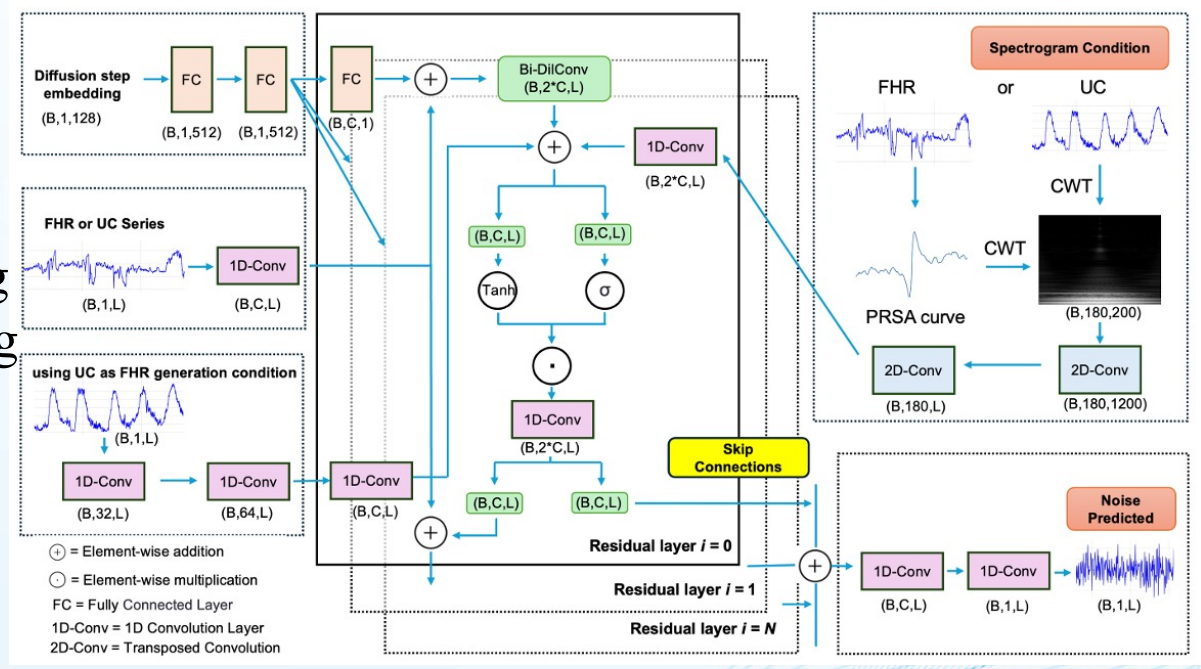
We conducted a visual inspection to assess the diversity of FHR signal samples generated by unconditional and conditional DDPM models.

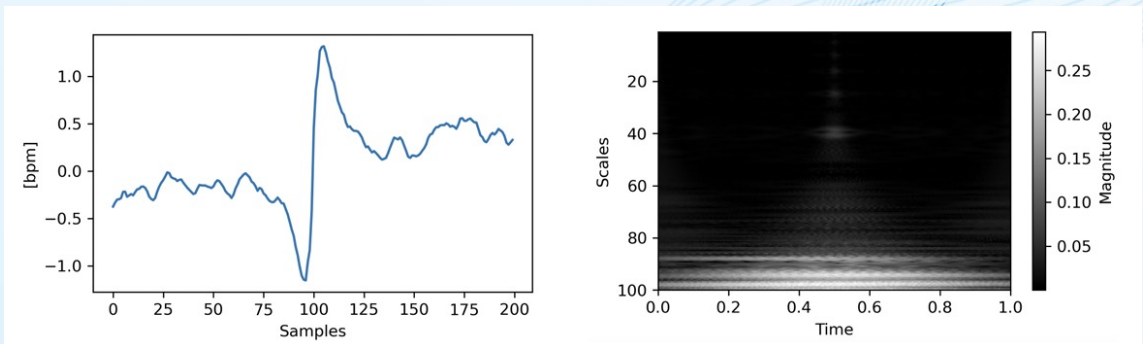


# 4 Conclusion

This study introduced a novel diffusion probabilistic model conditioned on PRSA spectrograms for synthesizing CTG signals. Our model captures essential physiological and pathological characteristics, generating high-fidelity and diverse data.




(a) PRSA curve of FHR series. (b) CWT of PRSA data.

# Thanks!